

## ☒ Easy Level (10 Questions)

1.

Which symbol is used to redirect output to a file in Bash?

- A) &
- B) |
- C) >
- D) #

**Answer: C**

☒ > redirects standard output.

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2.

Which command shows the exit status of the last executed command?

- A) \$?
- B) \$@
- C) \$!
- D) \$#

**Answer: A**

☒ \$? returns the exit status.

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3.

Which command enables debug mode in a Bash script?

- A) bash -debug
- B) bash -x
- C) bash -d
- D) bash -r

**Answer: B**

☒ bash -x script.sh prints each command as it executes.

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4.

**What does set -e do in Bash?**

- A) Enables echo
- B) Exits on error
- C) Evaluates expression
- D) Enables export

**Answer: B**

☒ Exits the script if any command fails.

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5.

**What is the correct syntax to assign a variable in Bash?**

- A) name = "linux"
- B) \$name = linux
- C) name="linux"
- D) let name=linux

**Answer: C**

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6.

**What is the default exit status of a successful Bash command?**

- A) 1
- B) 0
- C) -1
- D) 255

**Answer: B**

☒ Exit code 0 means success.

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7.

**What does the echo command do?**

- A) Prints input
- B) Reads file
- C) Starts a loop
- D) Compiles code

**Answer: A**

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8.

**Which loop runs while a condition is true?**

- A) until
- B) while
- C) do
- D) for

**Answer: B**

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9.

**Which operator is used to compare two strings in Bash?**

- A) ==
- B) eq
- C) !=
- D) =

**Answer: D**

☒ For strings, = or != is used in [ ].

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10.

**Which file test operator checks if a file exists?**

- A) -r
- B) -e
- C) -d
- D) -x

**Answer: B**

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## ☒ Medium Level (20 Questions)

11.

**What is the use of 2> in Bash?**

- A) Merge output
- B) Redirect stdin
- C) Redirect stderr
- D) Read second argument

**Answer: C**

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12.

**How do you append output to a file instead of overwriting?**

- A) >
- B) >>
- C) |
- D) &&

**Answer: B**

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13.

**Which option sets a variable as read-only in Bash?**

- A) readonly
- B) const
- C) static
- D) immutable

**Answer: A**

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14.

**What will this script print?**

bash

Copy code

```
x=5 if [ $x -eq 5 ]; then echo "Yes"; else echo "No"; fi
```

- A) Yes
- B) No
- C) Error
- D) 5

**Answer: A**

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15.

**Which command shows script execution line-by-line?**

- A) set +x
- B) set -x
- C) bash -n
- D) trap -x

**Answer: B**

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16.

**What does trap do in a script?**

- A) Loops indefinitely
- B) Captures signals
- C) Catches errors
- D) Repeats commands

**Answer: B**

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17.

**How to extract substring from variable in Bash?**

- A) \${var:1:4}
- B) \$var(1-4)
- C) var.sub(1,4)
- D) extract(var)

**Answer: A**

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18.

**What does this do?**

bash

Copy code

```
if [[ "$str" =~ ^[0-9]+$ ]]; then echo "Number"; fi
```

- A) Matches alphabet
- B) Matches digits only
- C) Matches anything
- D) Syntax error

**Answer: B**



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19.

**Which loop executes at least once before checking the condition?**

- A) for
- B) while
- C) until
- D) None of these

**Answer: D**

☒ Bash does not have a do-while construct like some languages.

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20.

**Which command reads user input in Bash?**

- A) get
- B) scan
- C) read
- D) fetch

**Answer: C**

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21.

**Which option shows all declared variables in Bash?**

- A) list

B) show  
C) declare -p  
D) env  
**Answer: C**

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22.

**What will `[[ 5 -eq 5 ]] && echo Yes || echo No` print?**

A) No  
B) Yes  
C) Error  
D) 5  
**Answer: B**

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23.

**Which test checks if string length is zero?**

A) -n  
B) -z  
C) -e  
D) -s  
**Answer: B**

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24.

Which is NOT a valid Bash conditional?

A) `[[ $x -eq 5 ]]`

B) `[ $x -ne 10 ]`

C) `(( x < 10 ))`

D) `if ($x == 5)`

Answer: D

❌ Incorrect syntax for if in Bash.

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25.

What does this command do?

bash

Copy code

```
echo "file" > /dev/null
```

A) Deletes the file

- B) Silently discards output
- C) Writes to file
- D) Outputs to terminal

**Answer: B**

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26.

**Which redirect merges stderr and stdout?**

- A) 2>1
- B) >2&1
- C) 2>&1
- D) 1>&2

**Answer: C**

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27.

**What is the purpose of break in a loop?**

- A) Pause
- B) Restart
- C) Exit loop
- D) Skip iteration

**Answer: C**

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28.

**What will expr 5 + 2 return?**

- A) 52
- B) 7
- C) 5 + 2
- D) Syntax error

**Answer: B**

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29.

**How do you create a case-insensitive match in `[[ =~ ]]`?**

- A) No way
- B) Use `(?i)` in regex
- C) Use `-i` flag
- D) Use `shopt`

**Answer: B**

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30.

**Which of these reads from a file line-by-line?**

- A)

bash

Copy code

```
while read line; do echo $line; done < file.txt
```

- B) cat file.txt
- C) head file.txt
- D) read file.txt

**Answer: A**

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## 🚧 Hard Level (10 Questions)

31.

**What does set -o pipefail do?**

- A) Ignores pipe errors
- B) Exits if any command in a pipeline fails
- C) Logs pipe output
- D) Enables logging

**Answer: B**

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32.

**How do you check if a variable is defined and not null?**

- A) `[[ -v var ]]`
- B) `[ -n "$var" ]`
- C) `[[ $var != "" ]]`
- D) All of the above

**Answer: D**

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33.

**Which expression matches a string with only letters?**

- A) `^[a-zA-Z]+$`
- B) `^\d+$`
- C) `^[0-9a-z]`
- D) `.*`

**Answer: A**

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34.

**What will this output?**

bash

Copy code

```
var="abc"; echo "${var^}"
```

- A) abc
- B) ABC
- C) Abc
- D) aBc

**Answer: C**

☒ `${var^}` capitalizes the first character.

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35.

**What is the output of this?**

bash

Copy code



`(( 5 > 3 && 2 < 1 )) && echo Yes || echo No`

- A) Yes
- B) No
- C) Error
- D) 5

**Answer: B**

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36.

**What is the default value of a variable if unset?**

- A) null
- B) error
- C) empty string
- D) Not defined

**Answer: D**

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37.

**How do you capture output of a command into a variable?**

- A) `var=$(command)`
- B) `var=command`

C) var <- command

D) var==command

**Answer: A**

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38.

**What happens if you use set -u and access an undefined variable?**

A) Nothing

B) Default value used

C) Script exits

D) Prints warning

**Answer: C**

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39.

**Which regex matches "file1", "file2", and "file3"?**

A) file[1-3]

B) file{1..3}

C) file(1|2|3)

D) file\d

**Answer: A**

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40.

What is the difference between `[[ ... ]]` and `[ ... ]`?

A) None

B) `[[ ]]` allows regex and advanced expressions

C) `[ ]` is for strings only

D) `[ ]` can't be used in if

Answer: B